

Final Project Report for Stats 170B

California Home Price Forecasting

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GitHub: <https://github.com/charlesgao5199/Stats170-ZIPL0CK>

1 Introduction and Problem Statement

California housing prices vary sharply across neighborhoods, and local price changes matter for buyers, renters, planners, and policy analysts trying to understand affordability pressure. Our project ended up as a forecasting pipeline for California ZIP Code Tabulation Areas (ZCTAs): for each ZCTA-year record, we use information observed in the current year to predict the next year’s median sale price. This is a supervised regression problem, with the final holdout using 2023 features to predict 2024 prices.

The project combines housing-market data with demographic, school-quality, crime, amenity, and local minimum-wage data. This supports two questions. The prediction question is whether public ZCTA-level data can produce useful one-year-ahead forecasts. The interpretation question is which feature groups appear most useful for prediction, while recognizing that feature importance is not causal evidence.

The final models compare Ridge regression, Random Forest, and XGBoost on the same selected feature set. Ridge is an interpretable regularized linear baseline. Random Forest and XGBoost are tree-ensemble methods that are widely used for structured tabular prediction because they can capture nonlinear thresholds and interactions [1, 4]. Prior work on neighborhood change also motivates the use of local business and amenity indicators; for example, Glaeser, Kim, and Luca [5] show that business activity data can help measure and forecast neighborhood change.

On the final 2023 holdout, Ridge had the lowest mean absolute percentage error (MAPE) at 6.24%, Random Forest had the highest dollar-scale R^2 at 0.966, and XGBoost performed similarly with 6.80% MAPE and 0.963 dollar-scale R^2 . The main result is that all three models forecasted 2024 ZCTA-level median sale prices reasonably well, but added model complexity did not materially improve average percentage error. The project should therefore be read as a forecasting and model-comparison study, not as a causal study of what makes housing prices change.

2 Data Sets

The final dataset is a California ZCTA-year panel. Each row represents one ZCTA in one year, and the response is constructed from the following year’s median sale price relative to the current year’s median sale price. The final phase-2 table has 19,570 rows, 213 columns, covers 2012 through 2024, and includes 1,578 unique ZCTAs. The selected modeling schema contains 54 predictors: numeric measures, categorical geography/assessment fields, and boolean coverage or imputation indicators. After target construction, feature selection, and train-test splitting, the model-ready training and cross-validation set contains 12,530 rows from 2012 through 2022. The final holdout contains 1,138 rows from 2023, which are evaluated against 2024 median sale prices.

Table 1: Main source groups and their roles in the final modeling panel.

Source group	Examples of variables used	Role in project
Redfin listings [6]	Median sale price, inventory, months of supply, homes sold, sale-to-list ratio, off-market in two weeks	Current market conditions and recent momentum
ACS demographics [8, 10]	Population, median household income, education, age, race/ethnicity, nativity	Local socioeconomic context
School quality [2]	Number of schools, standardized performance measures, assessment regime indicators	Local education context
Crime estimates [3]	Property and violent crime estimates and rates	Neighborhood safety context
Amenities [9]	Coffee shops, restaurants, parks, fitness, total amenity rates	Local amenity and business activity context
Minimum wage [7]	State/local wage levels and imputation indicators	Local labor-market policy context

The most important data-management issue was geographic alignment. Redfin housing data is reported by ZIP code, while Census data is reported by ZCTA. ZIP codes are mail-delivery routes, while ZCTAs are Census-created geographic areas designed to approximate ZIP code areas. Because they do not always match one-to-one, we standardized records to a common `zcta5` key before joining data sources. Figure 1 shows the ZIP-to-ZCTA match status for Orange County as an example of why this alignment step matters.

Orange County ZIP-ZCTA Match Status

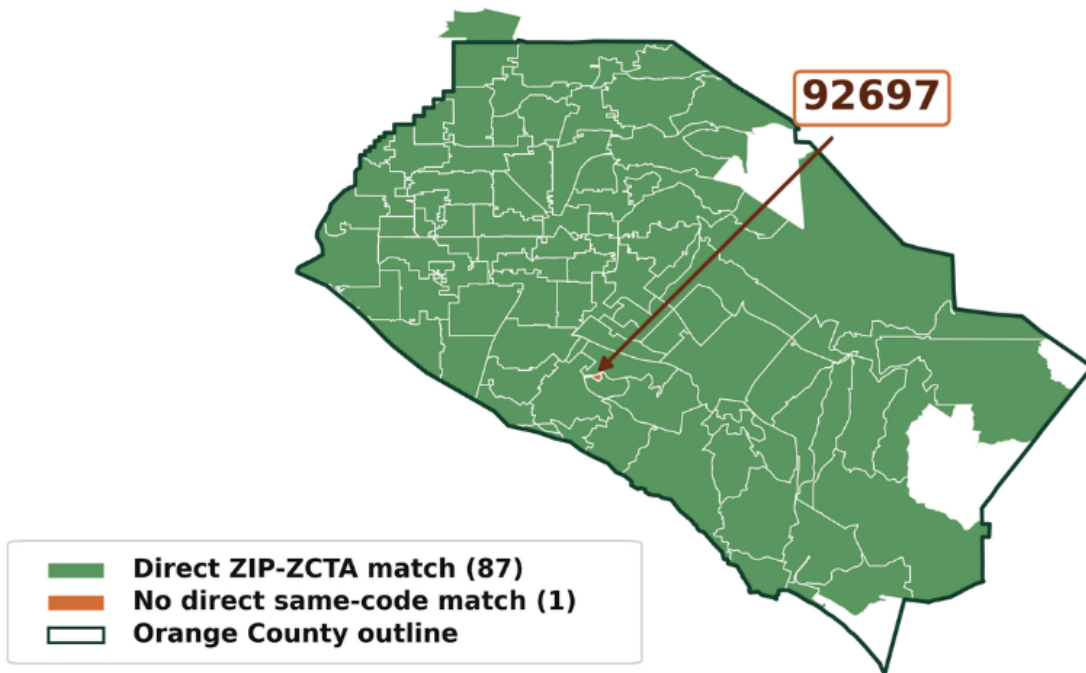


Figure 1: Orange County ZIP-to-ZCTA match status. Most Orange County ZIPs have direct same-code ZCTA matches, but some require spatial joining or special handling.

Other data-quality issues came from differences in time coverage and source granularity. School data has a transition gap around the change from STAR/CST to CAASPP testing and a COVID-era testing gap in 2020. Minimum-wage ordinances are defined at city, county, or state levels, so they had to be mapped into ZCTA-level records using geographic assignment rules. Crime and amenity records also required rate calculations so areas of different population sizes could be compared more fairly. These issues are why the final table keeps coverage, imputation, and source-availability fields rather than treating all missing values as equivalent.

Table 2 shows selected columns from the ZCTA-year modeling panel for ZCTA 92614 from 2012 through 2024. It is not the full modeling table; it displays a small subset of interpretable features to illustrate the panel structure. The full dataset contains many additional housing, demographic, school, crime, amenity, wage, coverage, and imputation variables. The 2020–2021 school counts and 2024 amenity count are carried forward from adjacent observed records so the example panel remains complete.

Table 2: Selected columns from example ZCTA-year records for ZCTA 92614. The full modeling table contains additional features not shown here.

ZCTA5	Year	Total population	Median sale price	Median income	Number of schools	Crime	Amenities
92614	2012	24,538	\$454,908	\$95,733	5	594	31
92614	2013	25,131	\$566,969	\$92,989	5	547	38
92614	2014	25,657	\$555,629	\$95,984	5	499	35
92614	2015	26,046	\$610,066	\$93,918	5	623	40
92614	2016	25,930	\$588,321	\$92,438	5	599	46
92614	2017	25,271	\$632,501	\$93,410	5	581	46
92614	2018	26,676	\$734,317	\$97,742	5	577	43
92614	2019	27,187	\$708,504	\$104,021	5	563	58
92614	2020	27,471	\$748,101	\$107,662	5	590	60
92614	2021	29,307	\$830,151	\$104,971	5	621	55
92614	2022	29,544	\$924,525	\$109,475	5	644	57
92614	2023	29,617	\$1,044,049	\$121,276	5	633	52
92614	2024	29,239	\$1,208,541	\$122,821	5	581	52

The final response variable is not raw total price. For ZCTA i in year t , we define the log price-change target as

$$y_{i,t} = \log \left(\frac{P_{i,t+1}}{P_{i,t}} \right),$$

where $P_{i,t}$ is the annual median sale price in ZCTA i during year t . This target behaves similarly to a percentage change and reduces the extent to which expensive ZCTAs dominate the learning problem. For final evaluation, predicted log changes are converted back into dollar-scale 2024 median sale prices so that MAPE, median absolute percentage error, and dollar-scale R^2 can be reported. (Response to first-draft feedback: the final response is a next-year log-change target and all final models use the same selected 54-feature schema.)

3 Overall Technical Approach

Figure 2 summarizes the data pipeline. Each public source was cleaned separately, standardized to California ZCTA-year records where possible, and then joined into one modeling table. The resulting panel was used to create the next-year log-change target, select predictors, train models, and evaluate forecasts on a final time-based holdout.

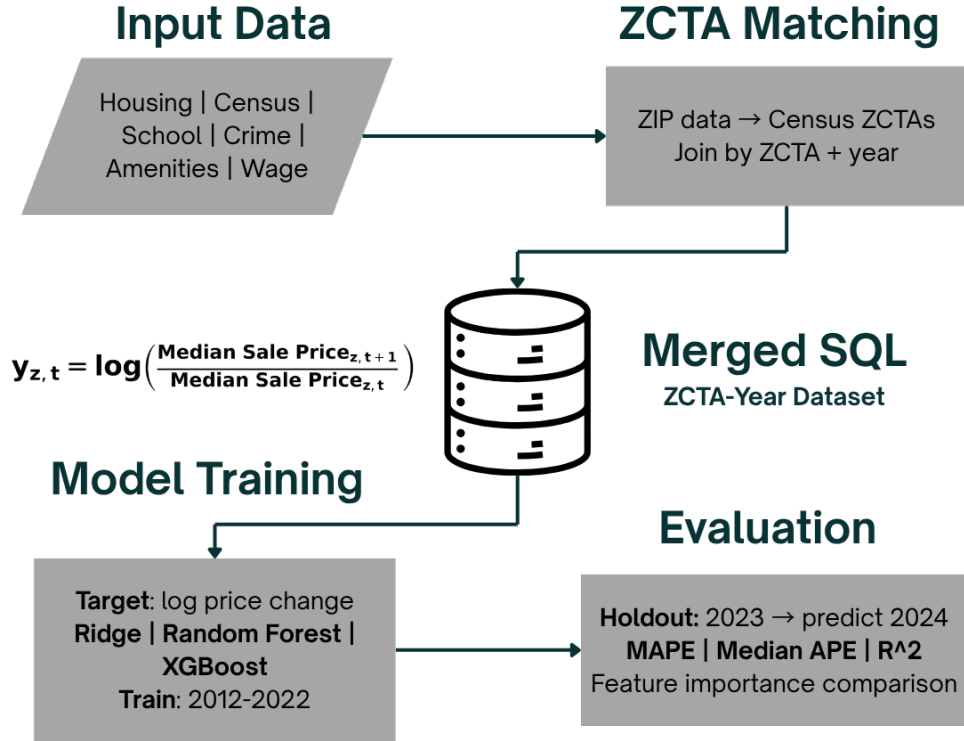


Figure 2: Project pipeline from public source data to model evaluation.

Data integration and feature selection. The feature-selection process produced 54 selected predictors used by all three final models. These include 12 housing/listings features, 25 demographic features, 6 school-quality features, 5 amenity features, 3 minimum-wage features, 2 crime features, and 1 other feature. Using the same selected feature set keeps the model comparison fair. We also removed variables that created leakage risk, excessive missingness, or unnecessary duplication.

Preprocessing and software. After feature selection, preprocessing handled numeric, categorical, and boolean variables. Numeric features were imputed and scaled for Ridge regression. Categorical variables such as metro region, assessment type, and county assignment quality were encoded. Boolean data-quality and availability indicators were kept so the models could account for missingness patterns and source coverage differences. The same train/test rows and selected predictors were used for Ridge, Random Forest, and XGBoost.

Table 3: Major software components used in the project.

Component	Functionality	Input/output
Project Python pipeline scripts	Clean source files, build feature tables, join ZCTA-year panel	Raw/source CSVs to model-ready CSVs
Model notebooks/scripts	Train Ridge, Random Forest, and XGBoost; generate metrics and figures	Selected train/test CSVs to predictions, metrics, plots
<code>pandas</code> , <code>numpy</code>	Tabular cleaning, joins, aggregation, feature construction	CSV tables to analysis tables
<code>scikit-learn</code> , XGBoost	Preprocessing, cross-validation, model fitting, evaluation	Feature matrices to fitted models and predictions
<code>matplotlib</code> /plotting code	Report and expo-ready visualizations	Metrics/predictions to PNG figures

Training, prediction, and validation. We used a time-respecting validation design rather than a random split. Training and cross-validation used observations from 2012 through 2022, and the final holdout used 2023 features to predict 2024 prices. Figure 3 shows the expanding time-blocked validation design. This approach better matches the real forecasting problem because the model must predict future prices from past data, and it avoids training on future-year information. Each model predicts next-year log change; for evaluation, that prediction is converted back to a dollar-scale median sale price using the

current-year median sale price.

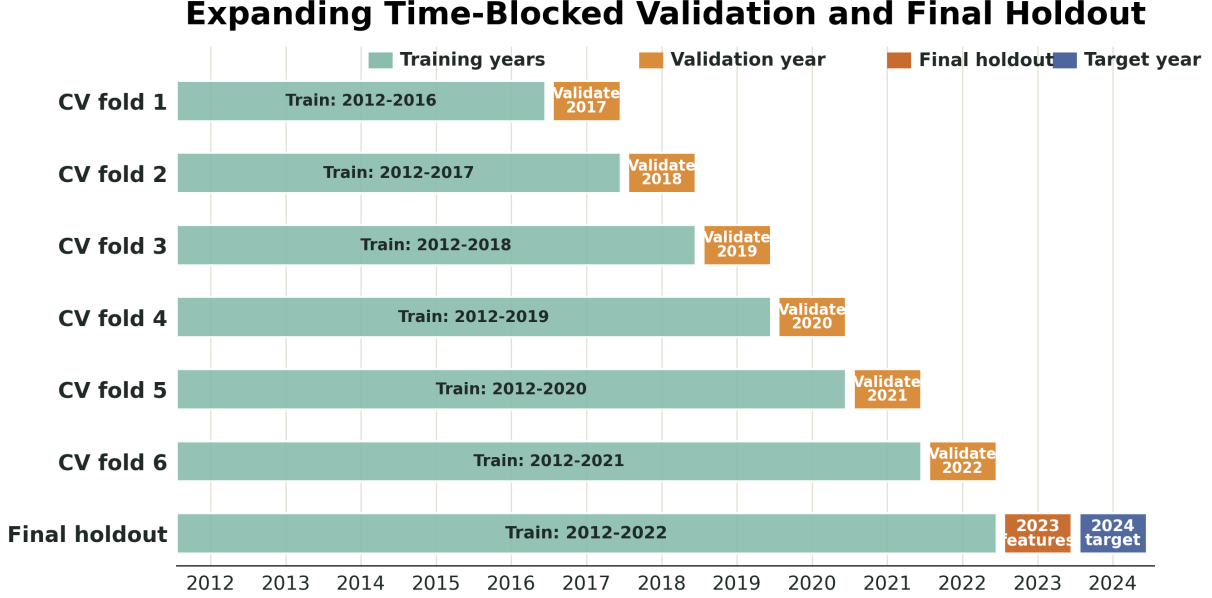


Figure 3: Expanding time-blocked validation and final 2023-to-2024 holdout.

Ridge regression is the linear baseline. It fits an additive linear relationship after preprocessing but uses an L2 penalty to shrink unstable coefficients. This is useful because the project has many correlated public-data features, and regularization reduces the risk of large unstable coefficients. Ridge is also easy to interpret, so a more complex model should improve performance enough to justify its extra complexity. Random Forest is an ensemble of decision trees [1]. Each tree is trained on a bootstrapped sample and considers random subsets of features at splits. Averaging across many trees reduces variance and lets the model capture nonlinear relationships, threshold effects, and feature interactions. Random Forest is a practical choice for tabular data with mixed feature types and nonlinear structure.

XGBoost is a boosted tree model [4]. Instead of growing independent trees and averaging them, XGBoost builds trees sequentially so later trees correct errors from earlier trees. It uses shrinkage, subsampling, and regularization to improve generalization. XGBoost is widely used for structured tabular prediction problems because it can model nonlinear patterns while controlling overfitting.

The main assumptions and diagnostics are tied to the forecasting setup. First, the time-blocked split checks whether models generalize forward in time. Second, predicted-versus-actual plots check whether forecasts are calibrated across the full price range. Third, percentage-error metrics prevent very expensive ZCTAs from dominating evaluation. Finally, feature importance is interpreted only as predictive signal within a fitted model, not as a causal ranking of neighborhood characteristics.

4 Evaluation

We evaluated the final models on the 2023 holdout, predicting 2024 median sale prices for 1,138 ZCTAs. After selecting the feature set and model specifications using the 2012–2022 training and validation period, each model was refit on the full 2012–2022 training set and evaluated once on the 2023 holdout rows. The primary metric is MAPE, because a \$50,000 error has different practical meaning in a low-price ZCTA than in a very high-price ZCTA. For actual price y_i , predicted price $\hat{y}_i$, and n holdout rows,

$$\text{MAPE} = \frac{100}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right|, \quad R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}.$$

We also report median absolute percentage error (median APE), which describes the typical error, and dollar-scale R^2 , which measures how well the reconstructed dollar predictions match actual 2024 median sale prices.

A good model should have low MAPE, low median APE, predicted-versus-actual points close to the diagonal line, and a clear advantage over the other candidate models. We treat Ridge as the baseline approach because it is the simplest regularized linear model and the easiest to interpret; Random Forest and XGBoost need to improve enough to justify their added complexity. We treat holdout MAPE below roughly 7% as practically useful. We also report log-change R^2 , but it is harder because year-to-year price growth is noisy; dollar-scale R^2 can be high because current price strongly anchors next-year price.

Table 4: Final 2023 holdout performance for predicting 2024 median sale price.

Model	MAPE (%)	Median APE (%)	R^2 log change	R^2 dollars
Ridge	6.24	4.18	-0.152	0.955
Random Forest	6.73	5.21	-0.205	0.966
XGBoost	6.80	5.41	-0.210	0.963

Table 4 shows the final holdout results. Ridge had the lowest MAPE and median APE, making it the strongest model by the headline percentage-error metrics. Relative to the Ridge baseline, Random Forest and XGBoost did not reduce average percentage error. Random Forest had the strongest dollar-scale R^2 , which means it tracked broad price levels especially well, but this did not change the headline model choice. Figure 4 shows Ridge predicted versus actual 2024 median sale prices using the full price range. The diagonal line represents perfect prediction. Most points lie close to the diagonal, so the model tracks the main price level well. The largest visible errors occur in unusual or high-price ZCTAs, which is consistent with the idea that annual public features do not fully capture local shocks. (Response to first-draft feedback: the predicted-versus-actual plot is kept full-size so the axes, points, and error pattern are readable.)

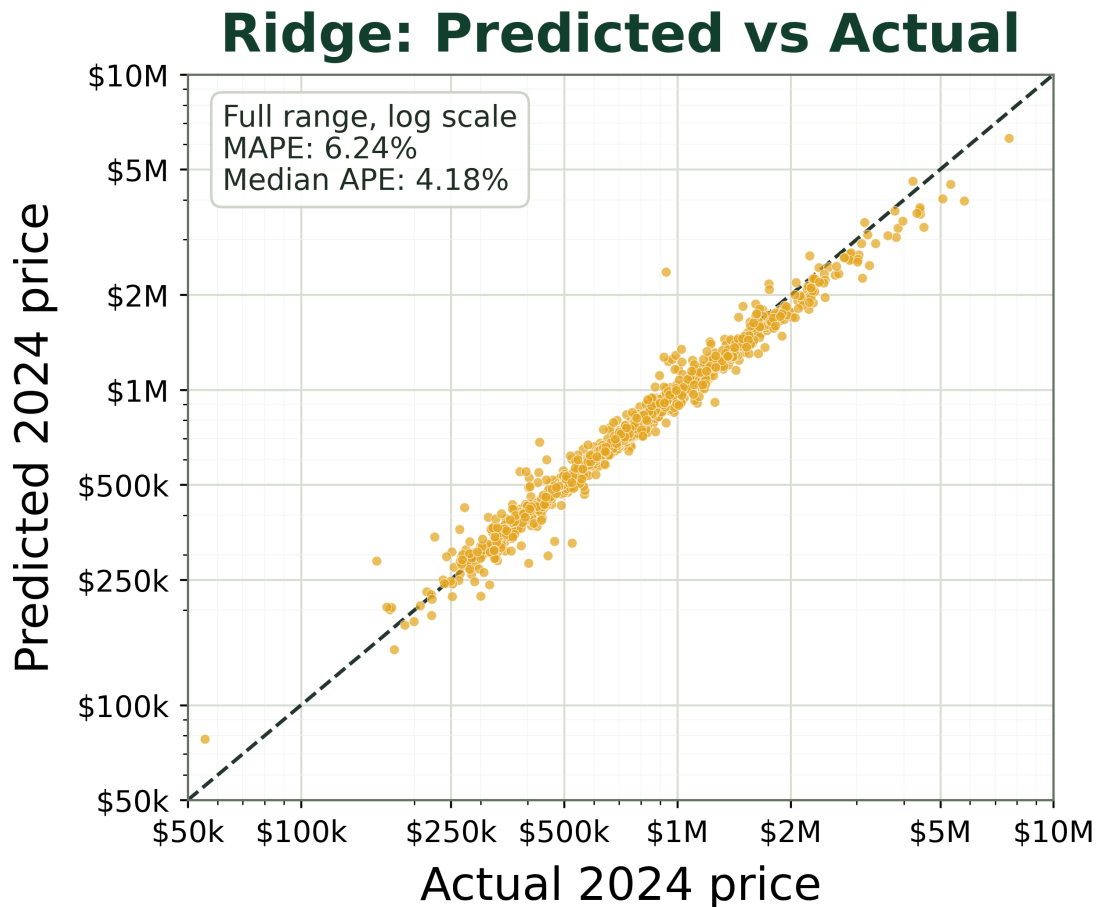


Figure 4: Ridge predicted versus actual 2024 median sale prices on the 2023 holdout.

Dollar-scale R^2 and percentage error tell different stories. Dollar-scale R^2 rewards anchored price levels,

while MAPE compares low- and high-price ZCTAs more fairly; this is why Ridge is strongest by our preferred metric even though Random Forest has a slightly higher dollar-scale R^2 .

Figure 5 shows the top four Ridge linear-regression coefficients by absolute standardized coefficient size. Ridge emphasizes assessment and school-coverage indicators, plus metro-region differences. These coefficients describe additive linear associations after preprocessing and regularization, so they are useful for interpretation but should not be read causally.

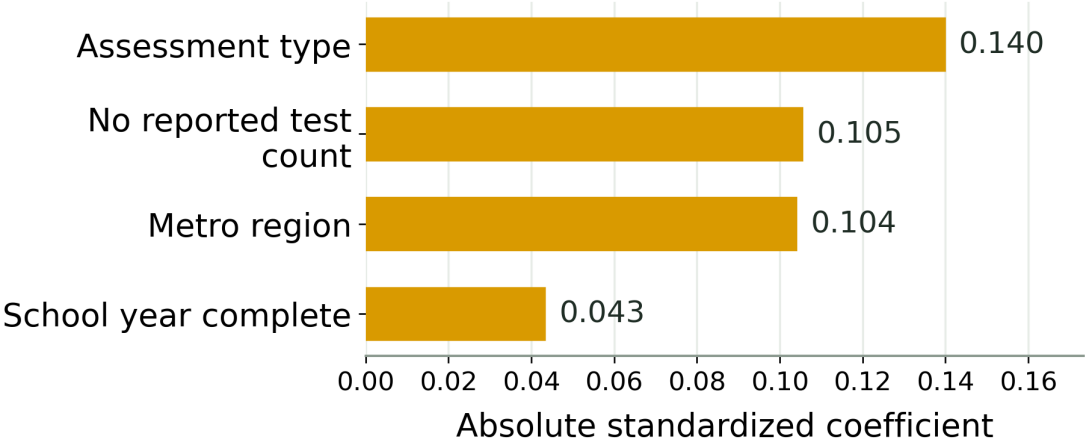


Figure 5: Ridge linear-regression top-four absolute standardized coefficients.

Figure 6 shows the top Random Forest permutation-importance features. Recent housing-market variables dominate: off-market in two weeks, months of supply, median sale price, and homes sold. Compared with Ridge, the tree model places more emphasis on direct market-state variables, while the linear model’s largest coefficients reflect assessment indicators and geography. This contrast suggests that the two model families use the same feature set differently: Ridge captures additive shifts, while Random Forest captures nonlinear market-condition patterns.

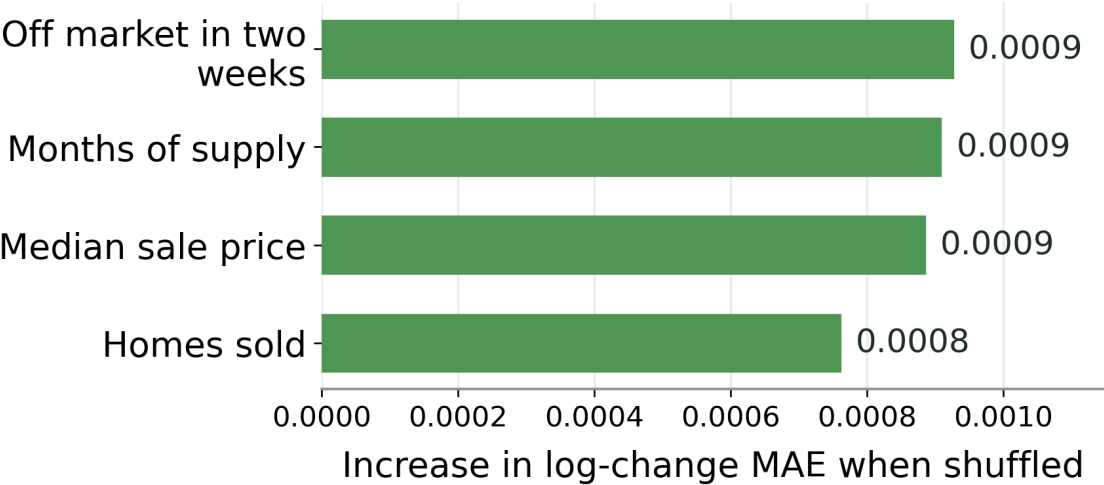


Figure 6: Random Forest top-four permutation importance features.

Overall, public ZCTA-level data forecast next-year median sale prices reasonably well, with average percentage errors around 6–7%. Ridge is the final recommended model because more complex nonlinear models did not beat it by the headline metric. The negative log-change R^2 values show that the models track anchored price levels better than they explain one-year growth.

5 Members Participation

Table 5: Team participation summary.

Task	Contributor split
1. Redfin housing data cleaning and annual listing features	Gary Zeng 100%
2. ACS demographic and ZCTA data processing	Charles Gao 100%
3. Amenities/business-pattern features	Anuj Patel 50%, Allen Lai 50%
4. School-quality feature construction and assessment-year handling	Charles Gao 40%, Gary Zeng 20%, Anuj Patel 20%, Allen Lai 20%
5. Crime and minimum-wage feature integration	Charles Gao 35%, Gary Zeng 25%, Anuj Patel 20%, Allen Lai 20%
6. Modeling, validation design, and final model comparison	Gary Zeng 25%, Charles Gao 25%, Anuj Patel 25%, Allen Lai 25%
7. Visualizations, report writing, poster, and final polishing	Gary Zeng 25%, Charles Gao 25%, Anuj Patel 25%, Allen Lai 25%

6 Discussion and Conclusion

- **Methods and algorithms.** Ridge was the strongest practical model because it had the lowest MAPE, was easy to explain, and made the coefficient directions transparent. Random Forest and XGBoost were useful for testing nonlinear structure and feature importance, but their extra complexity did not reduce average percentage error. A key limitation is that feature importance was model-specific: Ridge emphasized school-reporting and metro-region indicators, while Random Forest emphasized recent housing-market conditions.
- **Harder than expected.** Data integration was harder than model fitting. ZIP/ZCTA alignment, school assessment regime changes, missing source years, and local policy assignment all affected feature construction. These issues made the cleaned ZCTA-year panel and imputation rules as important as the choice of algorithm.
- **Surprising results.** The models forecasted 2024 price levels reasonably well, but one-year growth remained difficult. Recent median sale price and other market-state variables anchored predictions, while amenities, schools, crime, and wage features added context without fully capturing broader housing-market shocks.
- **Other lessons.** The project showed that temporal validation matters for housing data: random or geography-only splits can look easier than forward forecasting. Keeping one reproducible pipeline in BigQuery and Python also made it easier to compare Ridge, Random Forest, and XGBoost on the same 54-feature schema.
- **Future directions.** If we were extending this work in a research lab, we would add mortgage rates, inflation, unemployment, regional time-series features, neighboring-ZCTA features, region-specific or hierarchical models, prediction intervals, and shock-aware validation windows.

In its final form, the project produced a cleaned ZCTA-year pipeline, a selected 54-feature schema, comparable Ridge/Random Forest/XGBoost models, and a final holdout evaluation for 2024 prices. The main conclusion is that public ZCTA-level data can support useful next-year price forecasts, but major progress on growth forecasts likely requires stronger macroeconomic and regional time-series information.

References

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